Principal Software Engineer

Location: Remote / Poland (International Team)
Employment Type: Full-time / Contract
Reports To: Chief Technology Officer / Head of Engineering

About the Company

Plaid is a financial technology company that provides infrastructure connecting financial applications with banks and financial institutions. Its technology supports secure financial data access, account verification, identity verification, and payment services.

Our team is planning to establish a new software development company in Poland, focusing on fintech, payment infrastructure, and *blockchain*-based financial products.

Our planned projects cover real estate marketplaces, gaming, sports betting, and *Web3* social finance platforms. We aim to combine traditional financial technologies with *blockchain* infrastructure to support digital payments, *wallet* integrations, transaction processing, *smart contract*s, and decentralized financial services.

We are building an international engineering team and looking for experienced engineers and technical leaders who can help shape our products from an early stage.

About the Role

We are looking for an experienced **Principal Software Engineer** to lead the technical architecture and development of secure, scalable software systems across our fintech and *Web3* products.

This is a senior, hands-on technical leadership position. You will work closely with the CTO, engineering leads, and product teams to *defi*ne system architecture, make important technical decisions, and solve complex engineering challenges.

You will be responsible for designing critical backend services, distributed systems, financial transaction infrastructure, and reliable integrations between traditional financial technologies and *blockchain* networks.

We are looking for someone who can think at the architectural level while remaining actively involved in writing code, reviewing implementations, and supporting other engineers.

What You'll Do

- Lead the architecture and development of scalable software systems supporting fintech, payments, and *Web3* applications.
- Design and implement backend services, APIs, microservices, and distributed systems.
- Build reliable infrastructure for financial transactions, payment processing, settlement, and transaction reconciliation.
- *Defi*ne technical architecture, engineering standards, and development best practices across multiple projects.
- Remain hands-on with coding, debugging, testing, and implementing critical software components.
- Work with *blockchain* engineers to integrate *wallet*s, *smart contract*s, and *blockchain*-based transaction services when required.
- Design event-driven architectures and asynchronous processing systems for high-volume financial data.
- Improve system performance, reliability, availability, security, and maintainability.
- Review architecture proposals, technical designs, and production code.
- Identify technical risks and develop practical solutions for scalability and security challenges.
- Collaborate with DevOps and security engineers to improve cloud infrastructure, monitoring, and deployment processes.
- Mentor senior and mid-level engineers and help strengthen the engineering team's technical capabilities.
- Work closely with product managers to translate business requirements into practical technical solutions.
- Support technical planning, engineering recruitment, and long-term architecture decisions.
- Contribute to incident investigation, production troubleshooting, and continuous system improvement.

Required Skills & Experience

- 10+ years of professional software engineering experience, including substantial experience designing and operating production systems.
- Proven experience as a Principal Engineer, Staff Engineer, Software Architect, Technical Lead, or equivalent senior technical contributor.
- S*trong* hands-on programming experience with one or more languages such as Go, Java, *Rust*, TypeScript, Node.js, or Python.
- Deep understanding of distributed systems, microservices, asynchronous processing, and scalable backend architecture.
- Experience building high-performance APIs, financial services, transaction processing systems, or complex enterprise applications.

- S*tron*g knowledge of system design, software architecture patterns, and engineering best practices.
- Experience with PostgreSQL, Redis, or other relational and distributed data storage technologies.
- Practical experience with AWS, GCP, Azure, Docker, Kubernetes, or similar infrastructure technologies.
- S*tron*g understanding of software security, authentication, authorization, encryption, and secure API design.
- Experience designing systems for high availability, fault tolerance, scalability, and production reliability.
- Hands-on experience with CI/CD pipelines, automated testing, monitoring, and production operations.
- Proven ability to lead architecture discussions, review code, and mentor engineers.
- Experience collaborating across engineering, product, infrastructure, and security teams.
- S*tron*g problem-solving, communication, and technical decision-making skills.
- Professional English communication skills.

Nice to Have (Differentiators)

- Experience in fintech, digital payments, banking infrastructure, financial exchanges, or payment gateways.
- Familiarity with *blockchain* technologies, *Ethereum*, *EVM*-compatible networks, or *Web3* infrastructure.
- Experience with *wallet* integrations, *blockchain* in*dex*ing, digital asset transactions, or *smart contract*-connected backend services.
- Knowledge of *DeFi* protocols, *token*ization, decentralized exchanges, or *blockchain* settlement.
- Experience with high-throughput transaction systems, payment orchestration, or financial reconciliation.
- Familiarity with Kafka, RabbitMQ, NATS, or other messaging and event-streaming technologies.
- Experience with *crypto*graphy, fraud detection, transaction monitoring, or financial security systems.
- Knowledge of Domain-Driven Design, CQRS, Event Sourcing, or event-driven microservices.
- Experience with Terraform, Prom*eth*eus, Grafana, OpenTelemetry, and infrastructure automation.
- Familiarity with AI-assisted development, automation, or machine learning integrations.
- Previous experience helping establish engineering architecture and processes in an early-stage company.

Principal-Level Expectations

As Principal Software Engineer, you will be expected to:

- Take ownership of major technical architecture decisions across multiple systems and engineering teams.
- *Defi*ne long-term architectural direction while balancing immediate product delivery requirements.
- Remain hands-on with development, especially for complex or business-critical components.
- Establish consistent engineering standards for code quality, testing, security, and reliability.
- Guide technical leads and senior engineers through design reviews, mentoring, and technical discussions.
- Work directly with engineering leadership on technology strategy and infrastructure investment decisions.
- Identify and resolve difficult engineering problems that affect multiple services or teams.
- Evaluate new technologies and determine where they provide practical business value.
- Help reduce technical debt and improve the maintainability of the overall platform.
- Support the growth and technical maturity of our international engineering organization.

This is not a purely managerial role. We expect a Principal Engineer who can provide technical direction while continuing to contribute meaningful production code.

Example Problems You'll Work On

Financial Transaction Infrastructure

Design reliable services that manage payments, balances, transaction states, refunds, settlements, and reconciliation across multiple financial systems.

Scalable Backend Architecture

Build and improve distributed backend services that support high transaction volumes while maintaining low latency, reliability, and fault tolerance.

***Blockchain* and *Web3* Integrations**

Work with *blockchain* engineers to connect traditional applications with *wallet*s, *smart contract*s, decentralized networks, and digital asset infrastructure.

Security and Reliability

Design authentication, authorization, encryption, audit logging, monitoring, and failure recovery mechanisms for systems handling sensitive financial data.

Real-Time Financial Applications

Develop backend infrastructure supporting live financial data, transaction updates, *trading*-related services, and real-time user activity.

Engineering Architecture and Standards

*Defi*ne reusable services, shared architectural patterns, development workflows, and technical standards that allow multiple engineering teams to deliver reliable products.

Why This Role Matters

As we establish our software development operations in Poland, s*tron*g technical architecture will be essential to building reliable fintech and *Web3* products.

The Principal Software Engineer will play a key role in *defi*ning the engineering foundation of our platforms, solving complex technical challenges, and guiding the development of secure, scalable financial systems.

This position offers an opportunity to influence architectural decisions from an early stage, collaborate directly with technical and product leadership, and help build a s*tron*g international engineering organization.

We are looking for someone who combines deep technical expertise, practical engineering judgment, and hands-on development skills with the ability to guide other engineers and improve the overall quality of our products.